14.11 Stop Line and Crosswalk Marking – Right Lanes – Multilane Roadway – Short and Long Duration

Purpose:

This layout shows the typical setup for marking stop lines and crosswalks in right lanes at an intersection.

For multilane intersections, this is usually done in stages, with the inner lane marking done separately from the outer lane marking to minimize disruption to traffic.

If the intersection requiring work is not depicted in the diagram, see the layouts in *Section 11: Traffic Control Layouts – Intersections.*

Standard:

- For short-duration work, a Crew Working Ahead C-004 sign is required.
- For long-duration work, a Construction Ahead C-018-1A sign is required.
- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab is required in advance of a second Right Lane Closed Ahead C-130-R sign.
- Flashing arrow boards (FABs) shall be used inside all lane closure tapers.
- If the speed limit is ≥ 70 km/h, a buffer space is required.

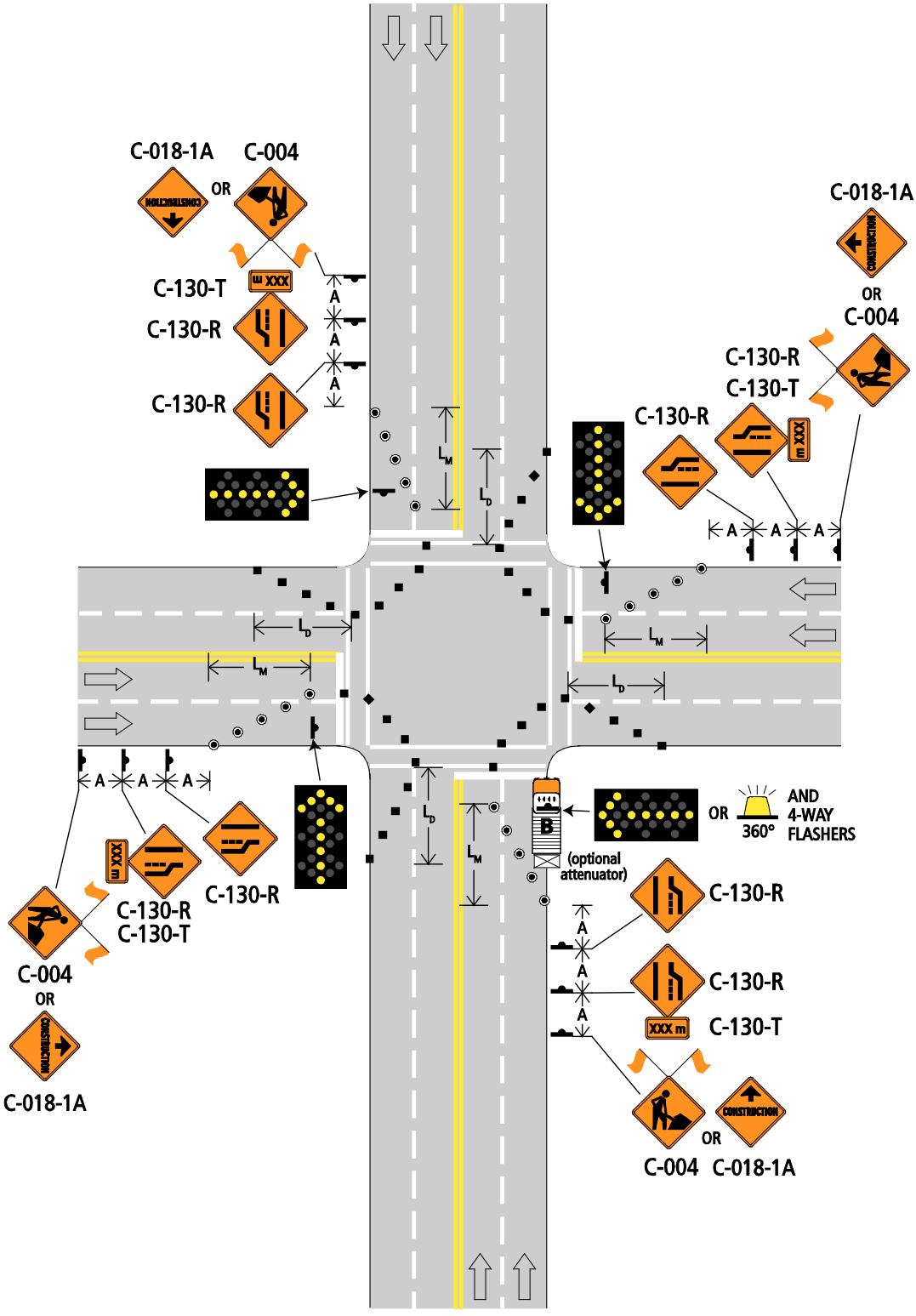
Guidance:

- A buffer vehicle with a 360-degree light and 4-way flashers should be used when workers are on the roadway.
- When a buffer space is provided, it may be less than the length specified in *Table B – Device Spacing* if space is limited and the adaptation is approved by the Road Authority.

Options:

- During periods of higher traffic volumes:
  - Left turns may be restricted.
  - The number of intersection legs with active work may be reduced.
- For low-volume, low-speed (≤ 60 km/h) roadways:
  - A Lane Closure Arrow C-053 sign may replace the flashing arrow board (FAB).
  - The advance lane drop sign and tab may be removed, and the Crew Working Ahead C-004 sign moved upstream by Table B Distance A.

Figure 14.11: Stop Line and Crosswalk Marking – Right Lanes – Multilane Roadway – Short and Long Duration



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PART B – TRAFFIC CONTROL

Section 15: Traffic Control Layouts – Surveying

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PART B – TRAFFIC CONTROL

Section 15: Traffic Control Layouts – Surveying

LEGEND			
	Flexible Drum		Paint Truck
	Tubular Marker		Escort Truck
	Cone		Chaser Vehicle
	Sign		Vehicle Mounted Rear Crash Attenuator
	Traffic Control Person		
	Work Activity Area		360° Flashing Light
	Work Truck		Portable Traffic Signal
	Shadow Vehicle		Barricade and Fencing
	Shadow Vehicle #1		Flashing Arrow Board (FAB)
	Shadow Vehicle #2		Flashing Arrow Board (FAB) in caution mode
	Buffer Vehicle		Dynamic Message Sign (DMS)

TABLE A – TAPER LENGTHS

Taper Types (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Merge Taper Length	L_M	35	55	160	190	210	230	250	280
Lane Shift Taper Length	L_L	30	50	80	100	110	120	130	140
Downstream Taper Length	L_D	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	L_S	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	L_T	30	60	160	190	210	230	250	280
Run-In Length on Centreline	L_R	40	50	60	60	70	80	90	100

TABLE B – DEVICE SPACING LENGTHS

Device Spacing (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Construction Sign Spacing	A	40	60	80	100	150	150	200	200
Buffer Space	B	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	R	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	C	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	D	10	10	30	30	40	40	40	50

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15.1 General Information - Surveying

1. Survey crews shall use extra caution because survey work areas often do not have the easily identifiable vehicles and equipment typically present at construction and maintenance work sites.
2. It may be advantageous for survey crew members to become qualified as Traffic Control Persons (see *Section 5: Traffic Control Persons)* so that they are authorized to control traffic when required. This will also enable them to supplement personnel who are working solely as TCPs.
3. All signs related to survey activities shall be removed or covered when no survey crew member is working on or adjacent to the roadway.
4. For traffic control layouts applicable to surveying on complex roadways, see:
    - *Section 7: Traffic Control Layouts – Two-Lane, Two-Way Roadways*
    - *Section 8: Traffic Control Layouts – Multilane Undivided Roadways*
    - *Section 9: Traffic Control Layouts – Multilane Divided Roadways*
    - *Section 11: Traffic Control Layouts – Intersections*

15.2 Surveying on Shoulder

Purpose:

This layout shows the typical setup for survey work occurring on the shoulder when both the instrument person and the instrument are off the travelled portion of the roadway.

Although the diagram depicts a two-lane, two-way roadway, this setup can also be used for multilane roadways.

Standard:

- A Survey Crew Working Ahead C-003 sign with flags is required in advance of the work area.
- For all speed limits, the work vehicle shall have a 360-degree flashing yellow light and 4-way flashers.

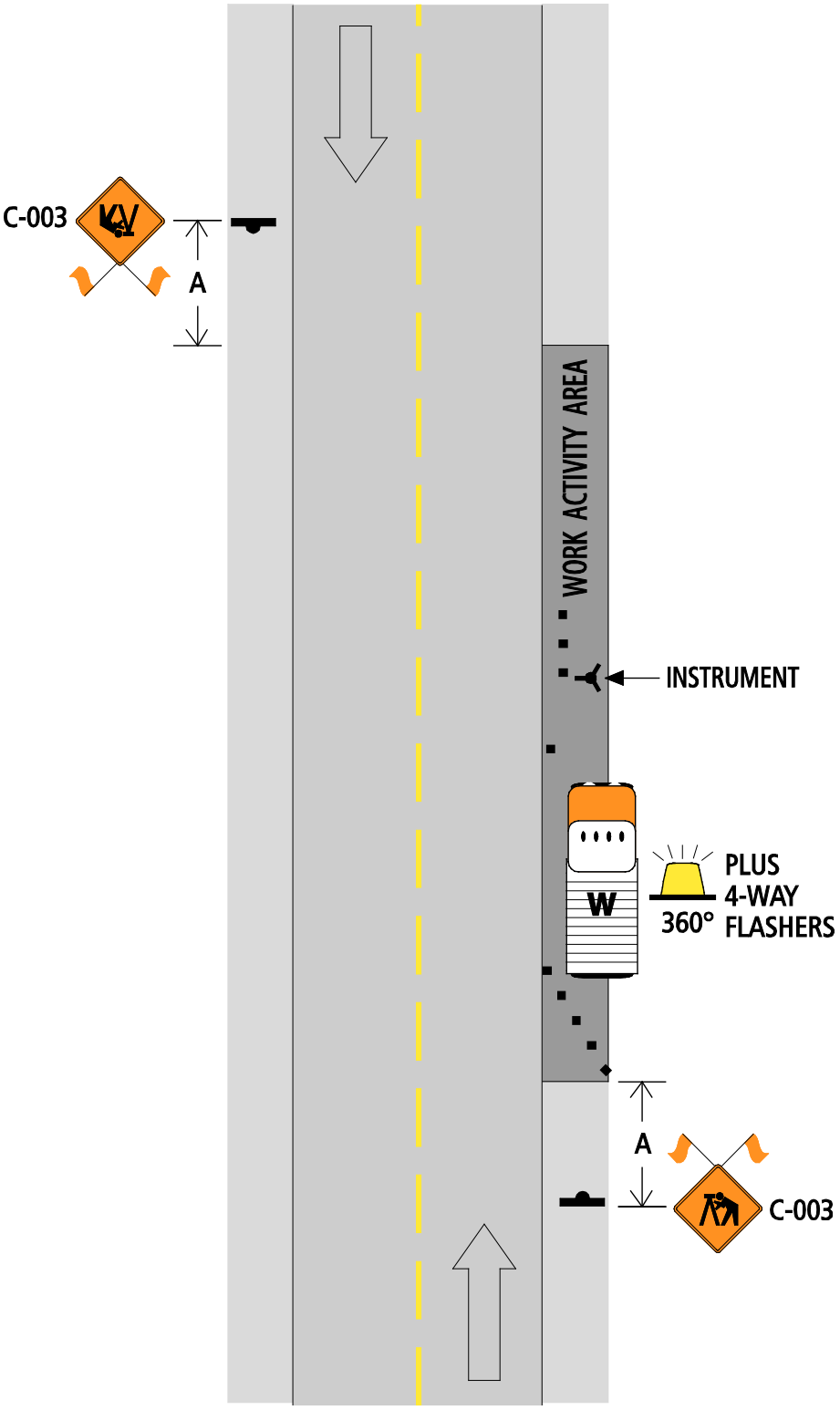
Guidance:

- For high-volume roads and speed limits ≥ 70 km/h, a buffer vehicle with a 360-degree flashing yellow light and 4-way flashers should also be used.
- If the roadway is a multilane divided roadway, the Survey Crew Working Ahead C-003 sign in the opposing direction should be omitted.

Options:

- A flashing arrow board (FAB) in caution mode may be used instead of the work vehicle with a 360-degree flashing yellow light and 4-way flashers.
- Where cyclists are regularly observed using the shoulder, a Share the Road W-132-1 sign may be used with an appropriate cycling hazard tab (see *Section 18: Traffic Control Layouts – Bicycle Lanes* for additional information).

Figure 15.2: Surveying on Shoulder



15.3 Surveying on Centreline

Purpose:

This layout shows the typical setup for surveying on centreline.

If passing traffic is a concern, other traffic control methods and layouts shall be used.

Standard:

- A Road Survey Ahead C-018-4 sign shall be used in advance of the general survey area. The distance between opposing C-018-4 signs shall not exceed 2 kilometres. The signs should be moved as the work progresses so that the work vehicles remain within the distance indicated on the signs.
- A Survey Crew Working Ahead C-003 sign with flags shall be used in advance of the work area.
- A shadow vehicle shall be used to provide advance warning to approaching vehicles.
- All vehicles shall be equipped with a 360-degree flashing light and 4-way flashers.
- If sufficient shoulder width (≥ 3.5 metres) is available for traffic to pass the survey crew on the right, a Pass this Side C-042-R sign shall be installed on the rear of the shadow vehicle. Otherwise, a Yield to Oncoming Traffic R-056-1 sign shall be used.

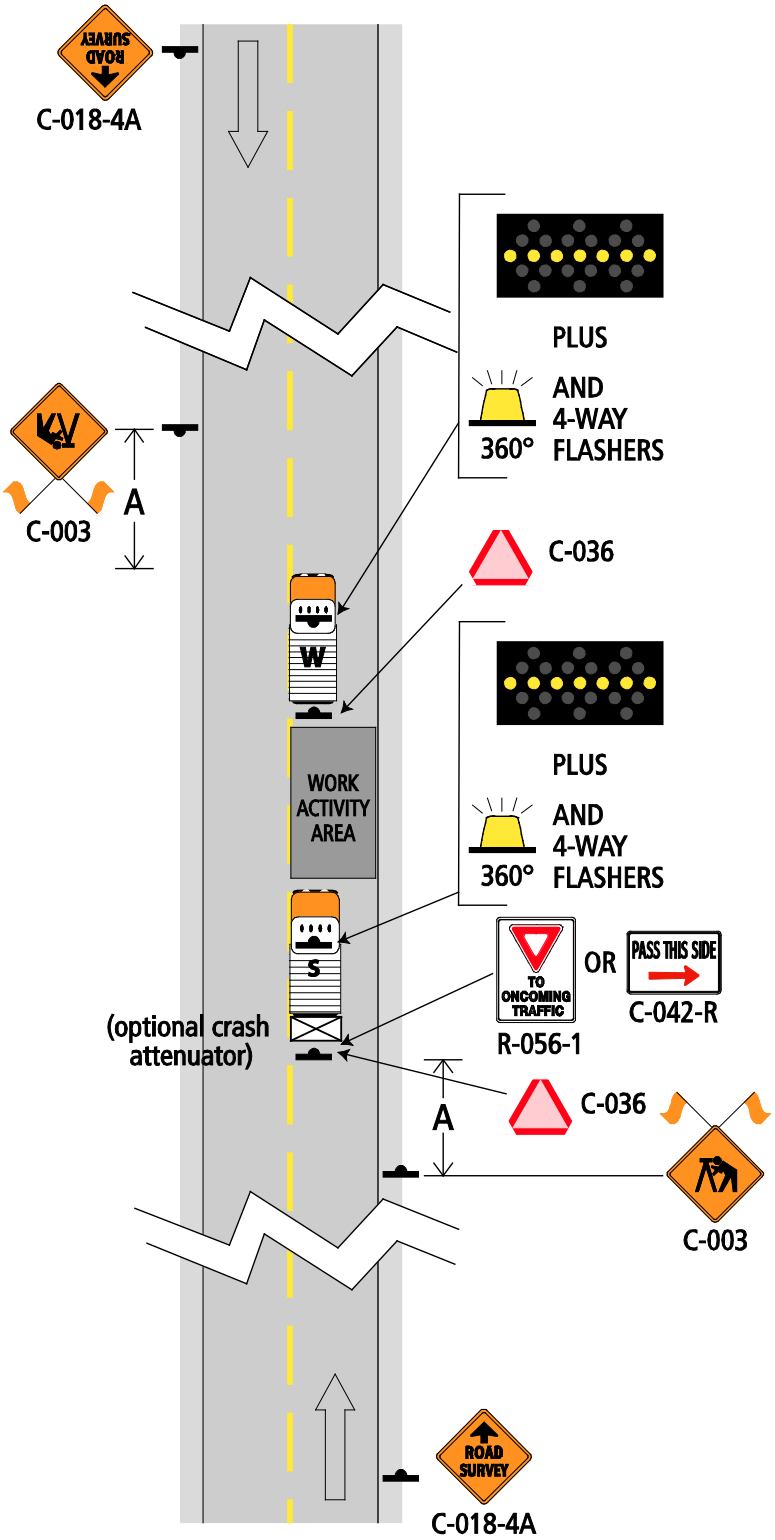
Guidance:

- A Survey Crew Working – Maximum Speed C-002-1 sign and a Thank You Resume Speed C-086-1 sign should be used only where conditions warrant (see *Section 2.4: Management of Speed* and *Section 5: Traffic Control Persons* for speed zone information). The distance between the opposing C-002-1 signs should not exceed 2 kilometres.

Options:

- The nature of the work activity area will vary, depending on the length and type of survey being completed. For example, a centreline survey may be completed using a rolling setup, in which case a more appropriate traffic control layout from the Manual should be applied.
- A spotter may be required to assist the instrument person by watching traffic.

Figure 15.3: Surveying on Centreline



15.4 Surveying in Intersections

Purpose:

This layout shows the typical setup for survey work occurring in an unsignalized intersection.

For more complex intersection layouts, see *Section 11: Traffic Control Layouts – Intersections.*

Standard:

- Survey Crew Working Ahead C-003 signs with flags shall be used in advance of the work area.
- A Survey Crew Working – Maximum Speed C-002-1 sign and a Thank You Resume Speed C-086-1 sign should be used only where conditions warrant (see *Section 2.4: Management of Speed* and *Section 5: Traffic Control Persons* for speed zone information).

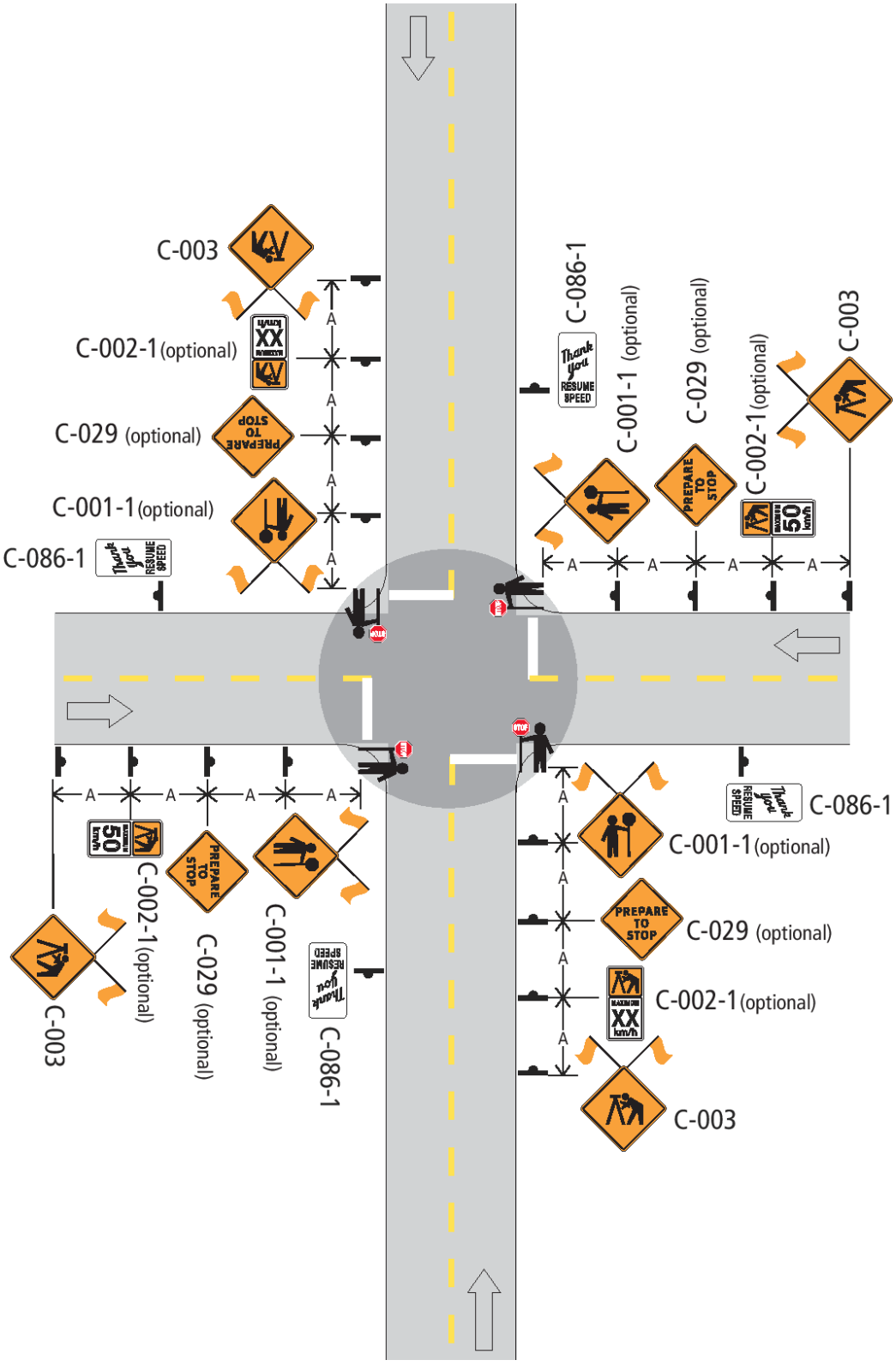
Guidance:

- If the Survey Crew Working – Maximum Speed C-002-1 sign is not required, the Survey Crew Working Ahead C-003 sign can be moved downstream by Table B Distance A.
- If no Temporary Speed Zone is used, the Work Zone Ends C-088 sign can be used instead of the Thank You Resume Speed C-086-1 sign.

Options:

- Traffic Control Persons are optional for this setup. If they are used, the Traffic Control Person Ahead C-001-1 sign and Prepare to Stop C-029 sign are required and the Flagger Ahead C-001-2 sign may be used for additional advance warning.
- When TCPs are directing traffic, the construction speed limit shall be ≤ 70 km/h.

Figure 15.4: Surveying in Intersections



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PART B – TRAFFIC CONTROL

Section 16: Traffic Control Layouts – Avalanche Control

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PART B– TRAFFIC CONTROL

Section 16: Traffic Control Layouts – Avalanche Control

LEGEND			
	Flexible Drum		Paint Truck
	Tubular Marker		Escort Truck
	Cone		Chaser Vehicle
	Sign		Vehicle Mounted Rear Crash Attenuator
	Traffic Control Person		
	Work Activity Area		360° Flashing Light
	Work Truck		Portable Traffic Signal
	Shadow Vehicle		Barricade and Fencing
	Shadow Vehicle #1		Flashing Arrow Board (FAB)
	Shadow Vehicle #2		Flashing Arrow Board (FAB) in caution mode
	Buffer Vehicle		Dynamic Message Sign (DMS)

TABLE A – TAPER LENGTHS

Taper Types (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Merge Taper Length	L_M	35	55	160	190	210	230	250	280
Lane Shift Taper Length	L_L	30	50	80	100	110	120	130	140
Downstream Taper Length	L_D	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	L_S	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	L_T	30	60	160	190	210	230	250	280
Run-In Length on Centreline	L_R	40	50	60	60	70	80	90	100

TABLE B – DEVICE SPACING LENGTHS

Device Spacing (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Construction Sign Spacing	A	40	60	80	100	150	150	200	200
Buffer Space	B	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	R	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	C	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	D	10	10	30	30	40	40	40	50

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16.1 General Information

Where snow avalanche conditions warrant, a highway may be closed at:

- locations with gates
- designated locations without gates
- other locations approved by the Ministry Avalanche Technician

Closure locations should be on a relatively level grade, be free of avalanche hazard, and have a turnaround capacity for large vehicles.

The traffic control layouts are considered the minimum standard. The associated text description of each figure highlights the key standards as well as guidance and options that can be considered by the user. The following information is typical on most layouts in this section:

Standard:

- For night operations, Type A yellow flashing lights are required on Avalanche Control C-057 signs, Prepare to Stop C-029 signs, and Traffic Control Person Ahead C-001-1 signs (when TCPs are used).
- Gate bases should be marked with Hazard W-154-L/R markers.

Guidance:

- When signs are not required—including those on gates—they should be removed, folded, or covered, and flags and flashing lights should be removed except for the flashing red lights installed on avalanche gates.
- On multilane divided roadways, and where space allows, the specified signing should be repeated in the median straight across from the shoulder signage.
- In addition, signage may be erected on the left shoulder where space allows, provided that it does not block sight lines or conflict with other signage.
- This work typically takes place during daylight hours. If the work will continue into the night, any TCP stations should be illuminated by overhead lighting.

Options:

- Additional advance signing—such as Avalanche Control C-057 signs or dynamic message signs (DMS)—may be used on curvilinear approaches or where it is anticipated that vehicle queues may extend past the standard signage layouts.

16.2 Road Closure for Avalanche Control – Gates without TCPs

Purpose:

This layout shows the typical setup where the highway must be closed for avalanche control, and the existing gates are used, without Traffic Control Persons.

Standard:

- A Stop R-001 sign and a Road Closed R-012 sign shall be clearly displayed on the gate as shown in the diagram.
- Type B red flashing lights shall be erected on the gate.
- The Avalanche Control C-057 sign and the Prepare to Stop C-029 sign shall have flags during the daytime.
- A Stop Ahead C-111 sign shall be positioned in advance of the closed gates.

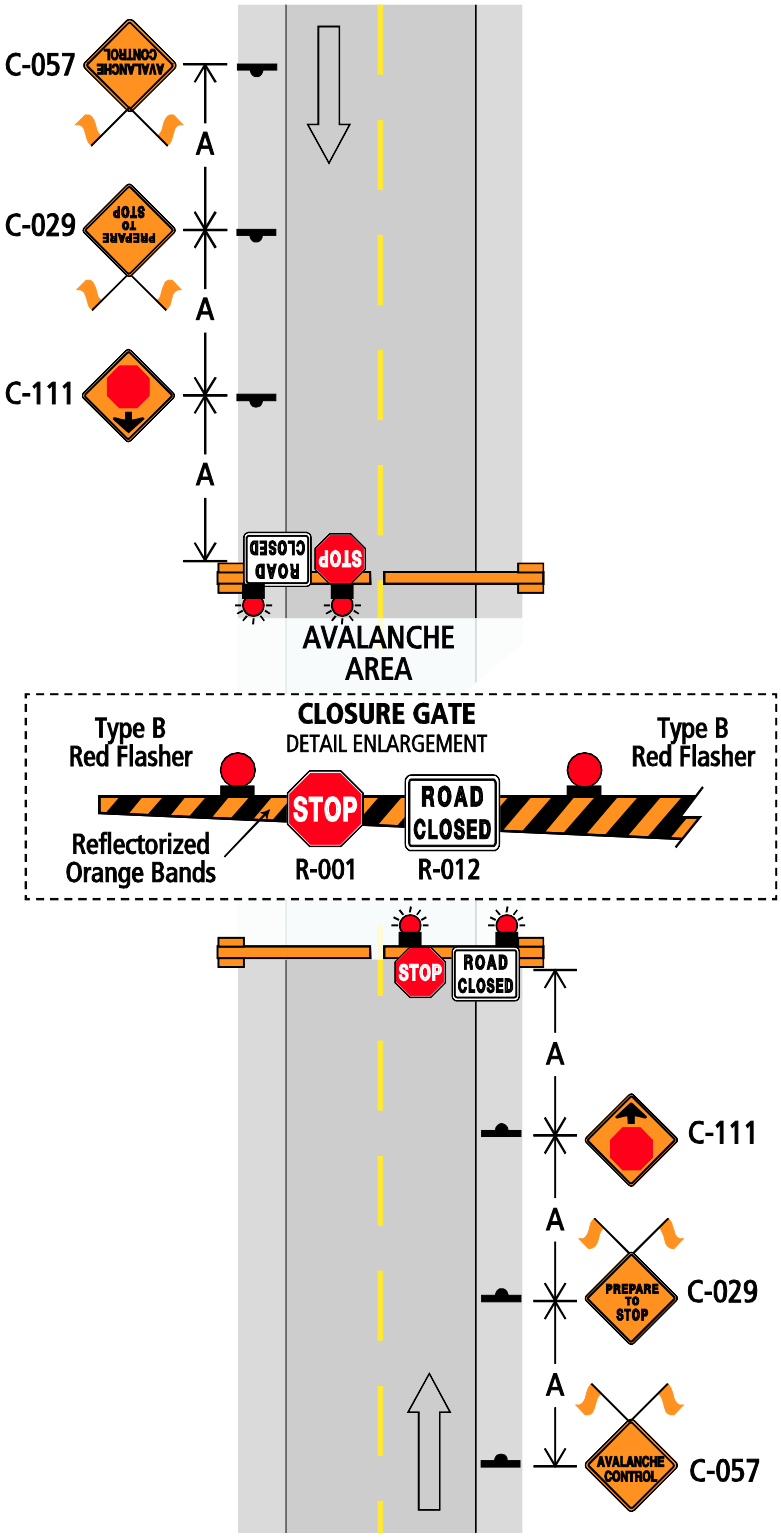
Guidance:

- If TCPs are not used, the avalanche barrier gates should be locked as authorized by the Ministry Avalanche Technician.
- Overhead permanent dynamic message signs (DMS) should be used whenever possible to inform travellers of the road closures.
- Additional portable dynamic message signs (DMS) may be used closer to the closure area to advise of closure periods, wait times, instructions for motorists in the waiting queue, and other details. See *Section 4.3.3: DMS Fundamentals* for sample messages and abbreviations.

Options:

- None at this time.

Figure 16.2: Road Closure for Avalanche Control – Gates without TCPs



16.3 Road Closure for Avalanche Control – Gates with TCPs

Purpose:

This layout shows the typical setup where the highway must be closed for avalanche control, and existing gates and Traffic Control Persons are both present.

In addition to directing traffic, TCPs may be useful for providing information to motorists about the closure, monitoring queue lengths, and adjusting signage as necessary.

Standard:

- A Stop R-001 sign and a Road Closed R-012 sign shall be clearly displayed on the gate as shown in the diagram.
- Type B red flashing lights shall be erected on the gate.
- The Avalanche Control C-057 sign, Prepare to Stop C-029 sign, and Traffic Control Person Ahead C-001-1 sign shall be positioned in advance of the TCP, and shall have flags during the daytime.

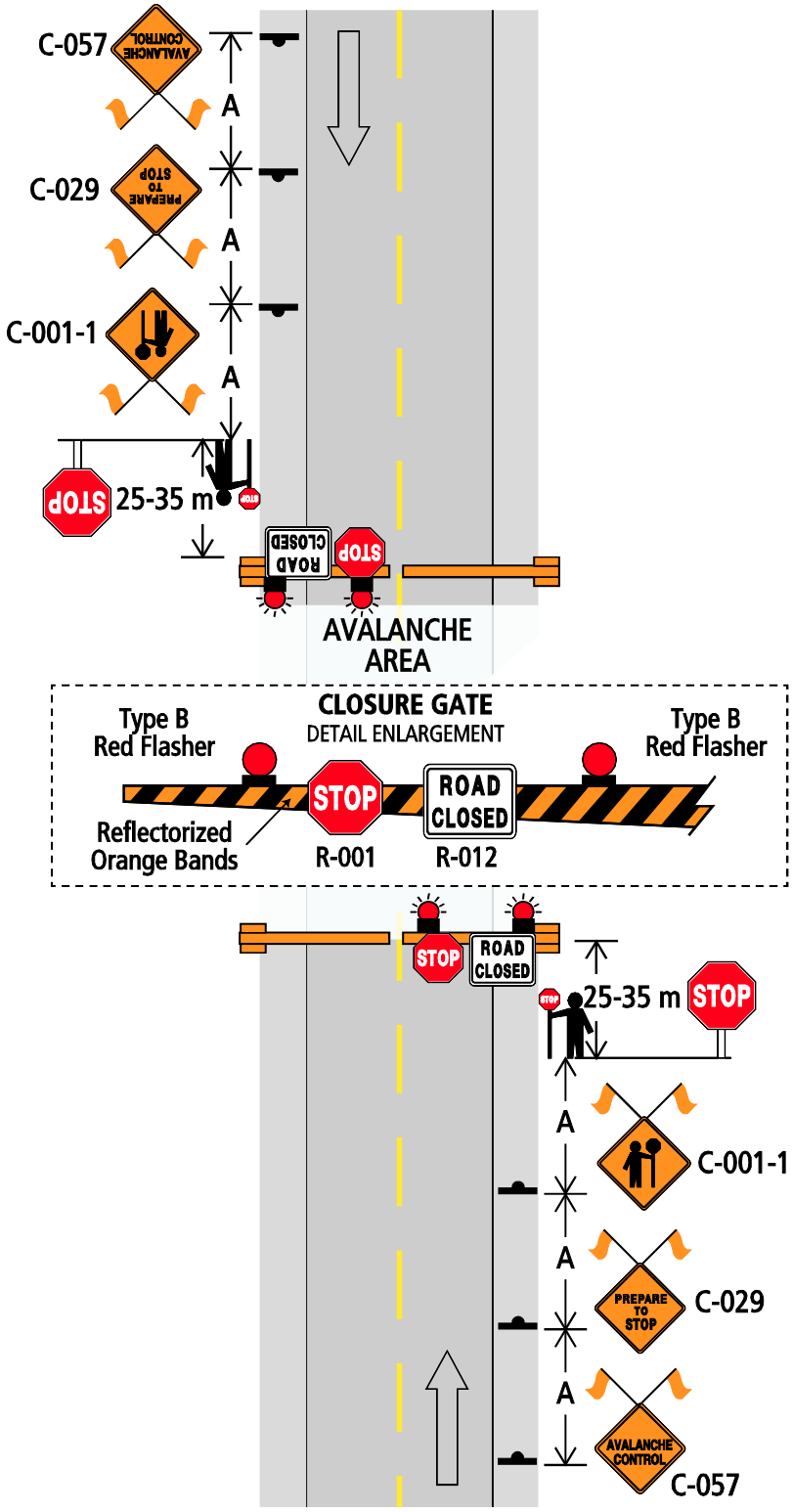
Guidance:

- Overhead permanent dynamic message signs (DMS) should be used whenever possible to inform travellers of the road closures.
- Additional portable dynamic message signs (DMS) may be used closer to the closure area to advise of closure periods, wait times, instructions for motorists in the waiting queue, and other details. See *Section 4.3.3: DMS Fundamentals* for sample messages and abbreviations.
- TCPs should monitor queue lengths, and adjust or add more advance warning signage as required.

Options:

- The Flagger Ahead C-001-2 sign or Prepare to Stop C-029 sign may be used for additional advance warning where TCPs are stopping traffic.
- When TCPs are directing traffic, the construction speed limit shall be ≤ 70 km/h.

Figure 16.3: Road Closure for Avalanche Control – Gates with TCPs



16.4 Road Closure for Avalanche Control – Barricades and TCPs

Purpose:

This layout shows the typical setup where the highway must be closed for avalanche control in a location where no gates are present.

When avalanches occur in non-gated areas, Ministry Avalanche Technicians determine the closure and traffic control requirements for using barricades and Traffic Control Persons.

In addition to directing traffic, TCPs may be useful for providing information to motorists about the closure, monitoring queue lengths, and adjusting signage as necessary.

Standard:

- A Stop R-001 sign and a Road Closed R-012 sign shall be clearly displayed on the barricade as shown in the diagram.
- Type B red flashing lights shall be erected on the barricade.
- The Avalanche Control C-057 sign, Prepare to Stop C-029 sign, and Traffic Control Person Ahead C-001-1 sign shall be positioned in advance of the TCP, and shall have flags during the daytime.

Guidance:

- Overhead permanent dynamic message signs (DMS) should be used whenever possible to inform travellers of the road closures.
- Additional portable dynamic message signs (DMS) may be used closer to the closure area to advise of closure periods, wait times, instructions for motorists in the waiting queue, and other details. See *Section 4.3.3: DMS Fundamentals* for sample messages and abbreviations.
- TCPs should monitor queue lengths, and adjust or add more advance warning signage as required.

Options:

- The Flagger Ahead C-001-2 sign or Prepare to Stop C-029 sign may be used for additional advance warning where TCPs are stopping traffic.
- When TCPs are directing traffic, the construction speed limit shall be ≤ 70 km/h.